# Problem Set 9

## E. Point Ordering

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

My Submissions

| # | When | Who | Problem | Lang | Verdict | Time | Memory |
|---|---|---|---|---|---|---|---|
| 387741742 | Aug/20/2026 12:23 UTC+10 | tianzheyu | A - Large Triangle | C++23 (GCC 14-64, msys2) | Accepted | 2750 ms | 59000 KB |

### Process

I implemented an $O(n^2 \log n)$ approach using a rotational sweep-line. I generated all possible base edges, sorted them by polar angle, and used binary search on a dynamically maintained permutation array to find the third vertex that matches the target area.

## Challenges and Overcoming

When maintaining the monotonic property of the points during the angular sweep, I ran into a bug where I accidentally swapped the actual `Point` objects in the main coordinate array instead of swapping their reference indices in the `cur_order` tracking array. I found it when I printed out the coordinates during the binary search step and noticed the structural integrity of my original point set was completely corrupted. Next time to prevent this bug, I will enforce strict separation between physical data and logical permutations by explicitly adding `const` modifiers to the base data vector, ensuring it remains read-only during the sweep line process.